Comparison of Four Types of Base Function Families and Research on GVZ Volatility Prediction

Introduction

The GVZ (Gold Volatility Index) is a gold volatility index released by the Chicago Board Options Exchange. It not only measures the market's expectations for future gold price fluctuations but is also an important indicator for financial risk management and derivative pricing. The GVZ prediction has a major impact on investment decisions and risk hedging as it can reflect the demand for hedging and market sentiment.

This study aims to select the optimal base function family by comparing the performance of four base function families (PWC, PWL, RBF, and Laplace) in time series modeling and prediction. It will predict the GVZ level from October to December in 2025 based on this. In the end, the study will utilise long-term historical data from 2008 to 2025 to analyse the fluctuation trend, identify key peaks and troughs, and combine the idea of smoothing and fitting to provide a basis for model construction. The structure of this article is as follows: The second part introduces the methods; the third part presents the empirical results; the fourth part summarises the findings and limitations.

Methodology

2.1 Model definition

This study employs the Least Squares Basis Function Regression (LBF):

$$y = \phi(x)^{\mathsf{T}}\beta + \varepsilon$$

x represents the month index, y represents the GVZ index, and ϕ(x) is the feature vector obtained after the basis function expansion. β is the parameter to be estimated. This model aims to capture the nonlinear relationship between time and volatility and provide a smooth estimation of the GVZ trend.

Given the value of k (the number of bases), β is estimated using the least squares method.

$$\hat{\beta} = (\Phi^{\mathsf{T}}\Phi)^{-1}\Phi^{\mathsf{T}}y$$

And make predictions using the formula $\hat{y} = \Phi\hat{\beta}$

2.2 Four types of basis functions

- PWC (Piecewise Constant): Can indicate structural changes, but prone to underfitting;
- PWL (Piecewise Linear): Can describe gradual transitions, but insensitive to extreme fluctuations;
- RBF (Radial Basis Function): Captures local patterns, flexible but prone to overfitting;
- Laplace (Laplace): Smooth, suitable for long-term trend structures.

2.3 Parameter Estimation and Numerical Issues

The β value is estimated using a closed-form solution. Considering the possible multicollinearity, scale differences, and overfitting, numerical stability control was implemented for Φ in the design. The RBF and Laplace have higher flexibility. If (k) is too large, it will result in excellent fitting of the training set but poor generalisation performance.

2.4 Hyperparameter selection and comparison scheme

Training set: 1–150; Validation set: 151–180; Test set: 181–208;

• Grid search over k ∈ [1, 30] to select the optimal k based on Validation MSE

• After parameter tuning for each group, compare Test MSE on the test set.

• Select parameters only on the validation set to avoid information leakage.

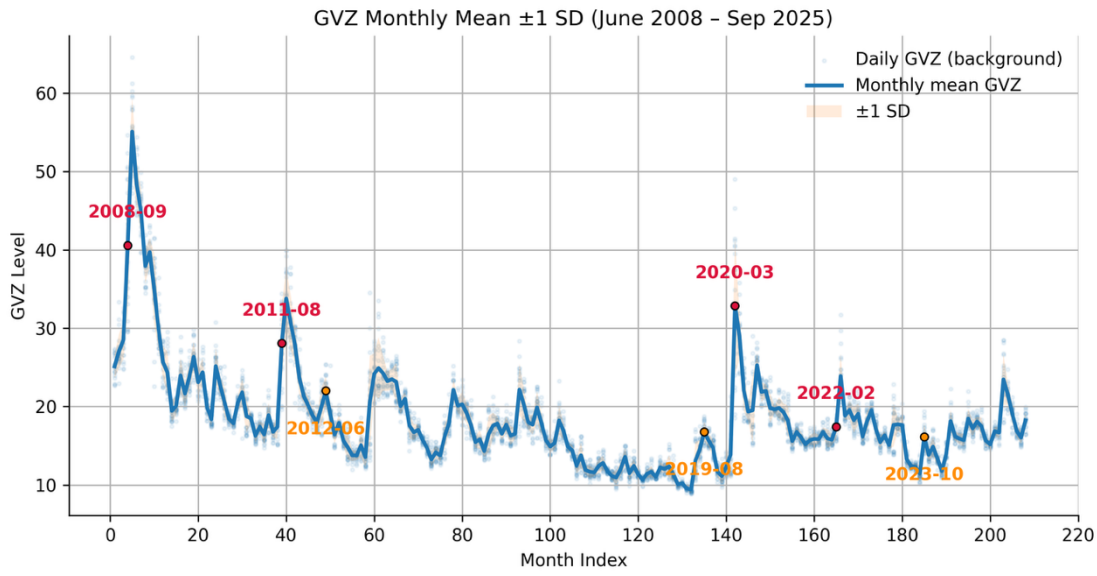


Figure 1 GVZ Monthly Mean ± 1 SD (June 2008 – September 2025).

The figure 1 shows the long-term fluctuation trend of GVZ from 2008 to 2025, mark the main peaks and troughs, corresponding to key market periods such as the global financial crisis in 2008 and the pandemic in 2020, and shows the low fluctuation range from 2013 to 2018, providing a basis for model smoothing fitting and base family selection.

Table 1

Optimal k and Validation MSE for Each Basis Function Family

	Basis	Best k	Validation MSE
0	laplace	1	7.048985
1	pwc	1	24.750505
2	pwl	15	16.399450
3	radial	2	7.652779

The table 1 shows that Laplace achieved the lowest error on the validation set, which initially displays that its fitting characteristics are more suitable for capturing the overall trend.

Empirical Study

3.1 Data Explanation and Initial Observations

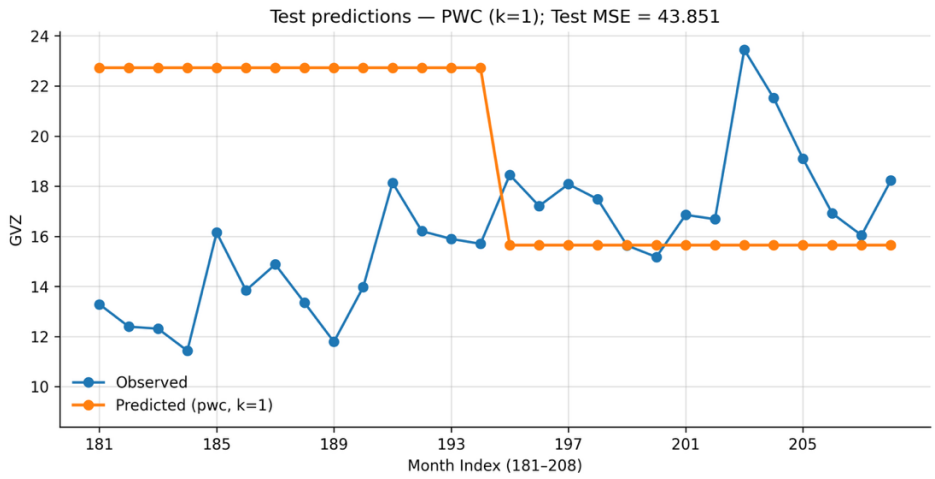
Data source: Federal Reserve Bank of St. Louis (FRED), USA.

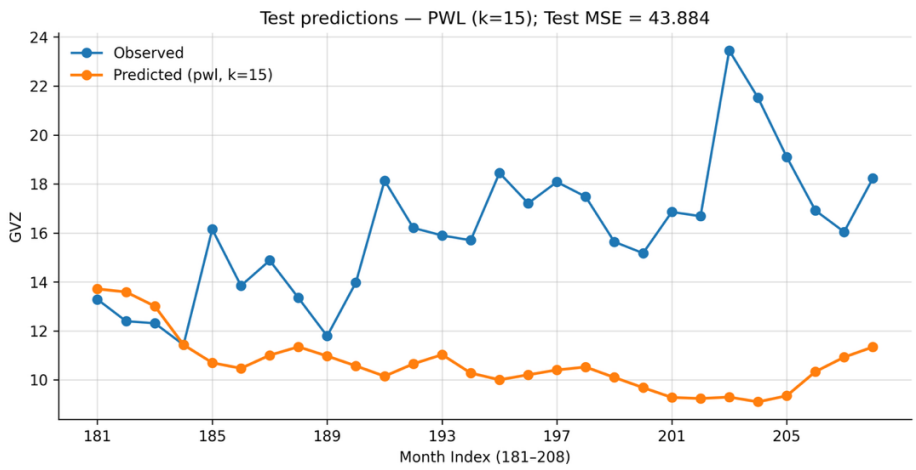
Time range: From January 2008 to September 2025, totaling 208 monthly observation values.

Preprocessing steps: Remove missing values, standardise time indices, and aggregate data by month.

Main observation: The fluctuation structure exhibits a distinct cyclical pattern, with the main peak periods corresponding to key market events such as the global financial crisis in 2008 and the COVID-19 pandemic in 2020.

3.2 Result Presentation





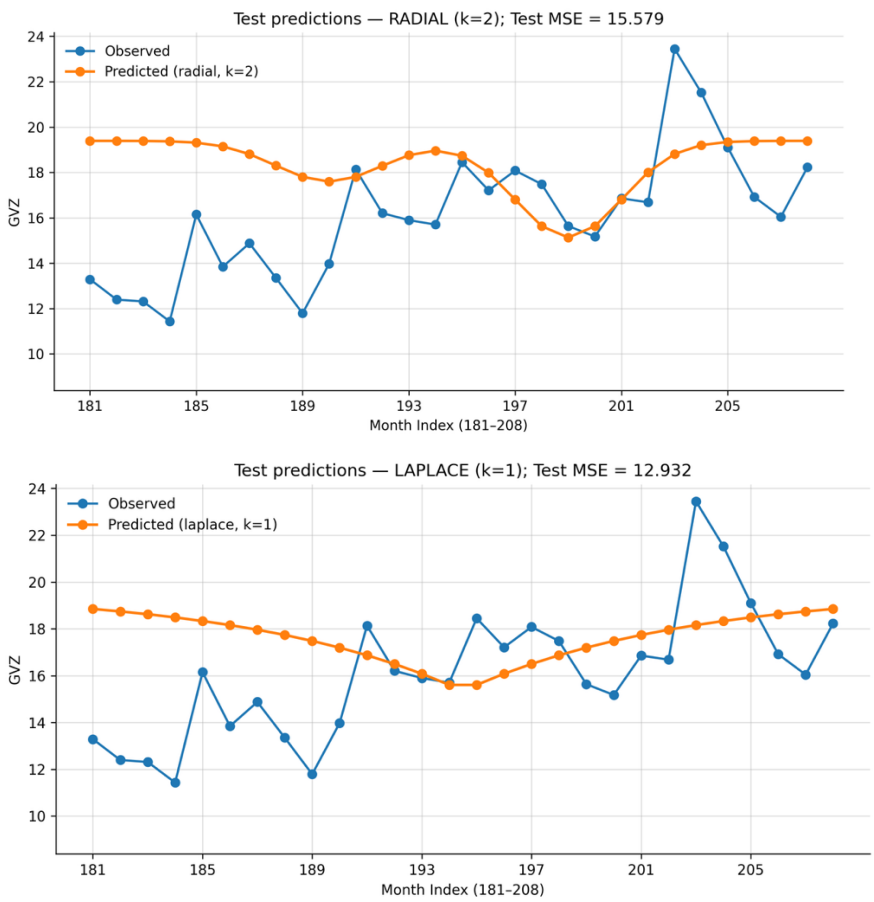


Figure 2 The prediction curves of the four models on the test set

The comparison results of the fitting curves of the four base families with the actual values show that the PWC and PWL models have an under-fitting problem and fail to capture the peak characteristics. The RBF model exhibits over-fitting locally and the curve fluctuates greatly. The Laplace model is the closest to the actual trend, is overall smooth and has a consistent trend. It shows the most stable performance.

Table 2

Test MSE of the Four Base Families

	Basis	Best k	Test MSE
0	laplace	1	12.932375
1	radial	2	15.578866
2	pwc	1	43.850529
3	pwl	15	43.883891

The Laplace basis family has the lowest error in the test set, followed by the RBF. The segmented model has weak generalisation ability and is difficult to reflect the overall trend.

3.3 Three-month forecast

Finally, the Laplace (k=1) model was selected for trend prediction. The prediction interval covers the Month Index from 209 to 211, corresponding to the period from October to December in 2025.

Table 3 *Predicted Values for the Next Three Months*

	Year-Month	Month Index	Forecast GVZ
0	2025-10	209	15.383328
1	2025-11	210	14.862524
2	2025-12	211	15.383328

The forecast indicates that GVZ will remain at a moderate fluctuating level of around 15 throughout the period from October to December in 2025. The overall trend is stable with no significant upward or downward indications.

Conclusion

This study compared four types of base function families. The results showed that the Laplace base function family performed the best on both the validation set and the test set. It can stably reflect the long-term fluctuation trend of GVZ. It is suitable for medium and long-term predictions. There are some limitations in the study. The model only uses the Month Index. It does not consider macroeconomic factors. Monthly data may mask short-term fluctuations. The model has not introduced regularisation and rolling validation. In the future, exogenous economic variables can be added to enhance the model's explanatory power. The regularisation method can also be introduced to improve the robustness. The model can also be combined with rolling prediction to improve its applicability and practical value in real-world scenarios.